

TrES-1b

R-0125

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_231011 · created 2026-07-16T05:45:34+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460228.6942 +/- 0.0026 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.113 +/- 0.011
Transit depth (Rp/Rs)^2	1.28 +/- 0.26 [%]
Orbital Inclination (inc)	89.99 +/- 0.74
Airmass coefficient 1 (a1)	1.275 +/- 0.011
Airmass coefficient 2 (a2)	-0.123 +/- 0.0033
Scatter in the residuals of the lightcurve fit is	0.82 %
Best Comparison Star	None
Optimal Aperture	4.23
Optimal Annulus	12.55
Transit Duration (day)	0.1015 +/- 0.0015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.113 ± 0.011 vs 0.1358 (deviation 2.07σ)
Duration fit vs published	0.1015 d vs 0.1042 d (deviation 1.78σ)
Mid-time O–C	6.9 min
Depth SNR	10.3
Residual scatter	0.82 %

Light curve

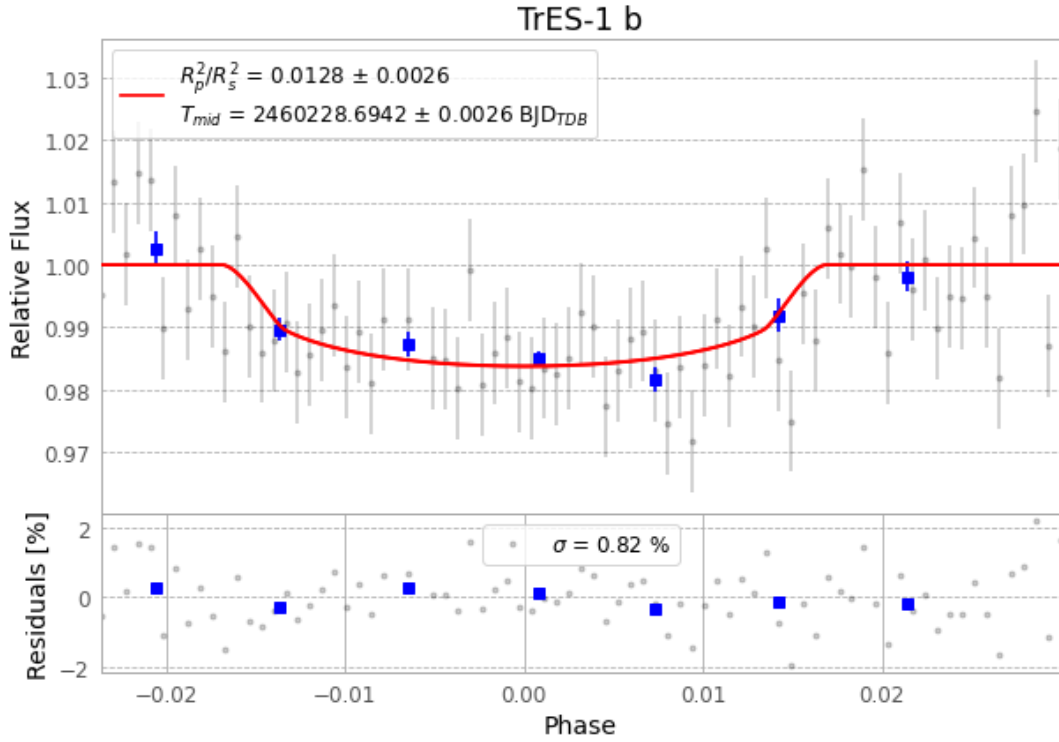


Figure 1 — FinalLightCurve_TrES-1 b_2023-10-10.png (SHA-256 2f7c67e859122368...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [266, 196], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7b9c7861bdde93dd...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 3ebc8ba

Data provenance

79 FITS frames (52 MB), TRES-1231011025415.FITS → TRES-1231011064815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-231011.log (0411557acc96...), FinalLightCurve_TrES-1 b_2023-10-10.png (2f7c67e85912...), FinalLightCurve_TrES-1 b_2023-10-10.csv (2eea2e3af8f3...)

Compute resources

Wall time 130.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 05:45Z.